

Parrot API-to-Backend Mapping

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Table 1: Parrot method mapping (core transforms, masking, and reductions).

Parrot (C++)	Parrot Python	CUDA C++ backend	cuda.compute backend
map(op)	map(op)	thrust::make_transform_iterator	iterators.TransformIterator
map2(arr, op)	map2(arr, op)	thrust::make_zip_iterator + thrust::make_transform_iterator	iterators.ZipIterator + iterators.TransformIterator
map2(scalar, op)	map2(scalar, op)	thrust::make_transform_iterator	clone + map(bound_op)
times(arg)	times(arg)	 map2(arg,mul)	 map2(arg,mul_op)
add(arg)	add(arg)	 map2(arg,add)	 map2(arg,add_op)
minus(arg)	minus(arg)	 map2(arg,minus)	 map2(arg,sub_op)
min(arg)	min(arg)	 map2(arg,min)	 map2(arg,min_op)
max(arg)	max(arg)	 map2(arg,max)	 map2(arg,max_op)
div(arg)	div(arg)	 map2(arg,div)	 map2(arg,div_op)
mod(arg)	mod(arg)	 map2(arg,mod)	 map2(arg,mod_op)
idiv(arg)	idiv(arg)	 map2(arg,idiv)	 map2(arg,idiv_op)
as<T>	astype(dtype)	 map(cast_functor)	 map(cast_op)
gte(arg)	gte(arg)	 map2(arg,gte)	 map2(arg,gte_op)
lte(arg)	lte(arg)	 map2(arg,lte)	 map2(arg,lte_op)
gt(arg)	gt(arg)	 map2(arg,gt)	 map2(arg,gt_op)
lt(arg)	lt(arg)	 map2(arg,lt)	 map2(arg,lt_op)
eq(arg)	eq(arg)	 map2(arg,eq)	 map2(arg,eq_op)
neq(arg)	neq(arg)	 map2(arg,neq)	 map2(arg,neq_op)
dbl()	double()	 map(double_functor)	 map(double_op)
half()	half()	 map(half_functor)	 map(half_op)
abs()	abs()	 map(abs_functor)	 map(abs_op)
log()	log()	 map(log_functor)	 map(log_op)
exp()	exp()	 map(exp_functor)	 map(exp_op)
neg()	neg()	 map(negate)	 map(neg_op)
odd()	odd()	 map(odd_functor)	 map(odd_op)
even()	even()	 map(even_functor)	 map(even_op)
sign()	sign()	 map(sign_functor)	 map(sign_op)
sqrt()	sqrt()	 map(sqrt_functor)	 map(sqrt_op)
sq()	sq()	 map(sq_functor)	 map(square_op)
rand()	rand()	zip(index,value) + transform random functor	ZipIterator + TransformIterator random op
map_adj(op)	map_adj(op)	thrust::make_zip_iterator + thrust::make_transform_iterator (zip_function)	PermutationIterator + ZipIterator + TransformIterator
differ()	differ()	 map_adj(neq)	 map_adj(neq_op)
deltas()	deltas()	 map_adj(delta)	 map_adj(delta_op)
keep(mask)	keep(mask)	mask-aware fusion_array constructor	lazy mask in ParrotArray (_mask)
filter(pred)	filter(predicate)	thrust::make_transform_iterator + keep 	algorithms.select
where()	where()	 range(size).keep(this)	 range(length).keep(self)
gather(indices)	gather(indices)	thrust::make_permutation_iterator	iterators.PermutationIterator
reduce<0>(init, op)	reduce(op, init, axis=0)	thrust::reduce	algorithms.reduce_into
reduce<1>(init, op)	reduce(op, init, axis=1)	 transpose().reduce<2>()	algorithms.segmented_reduce
reduce<2>(init, op)	reduce(op, init, axis=2)	thrustx::reduce_by_n	algorithms.segmented_reduce
sum<axis>()	sum(axis)	 reduce<axis>(0,+)	 reduce(+,0,axis)
prod<axis>()	prod(axis)	 reduce<axis>(1,*)	 reduce(*,1,axis)
maxr<axis>()	maxr/max	 reduce<axis>(init,max)	 reduce(max,init,axis)
minr<axis>()	minr/min	 reduce<axis>(init,min)	 reduce(min,init,axis)
any<axis>()	any(axis)	 reduce<axis>(init,logical_or)	 reduce(_reduce_or,...)
all<axis>()	all(axis)	 reduce<axis>(init,logical_and)	 reduce(_reduce_and,...)

Table 2: Parrot method mapping (scans, shape/rank operations, and products).

Parrot (C++)	Parrot Python	CUDA C++ backend	cuda.compute backend
scan<0>(op)	scan(op, axis=0)	thrust::inclusive_scan	algorithms.inclusive_scan
scan<1>(op)	scan(op, axis=1)	 transpose().scan<2>().transpose()	 (need scan_by_key)
scan<2>(op)	scan(op, axis=2)	thrust::inclusive_scan_by_key	 (need scan_by_key)
sums<axis>()	sums(axis)	 scan<axis>(+)	 _scan_to_array(+, 0)
prods<axis>()	prods(axis)	 scan<axis>(*)	 _scan_to_array(*, 1)
mins<axis>()	mins(axis)	 scan<axis>(min)	 _scan_to_array(min, init)
maxs<axis>()	maxs(axis)	 scan<axis>(max)	 _scan_to_array(max, init)
alls<axis>()	alls(axis)	 scan<axis>(logical_and)	 scan(logical_and,...)
anys<axis>()	anys(axis)	 scan<axis>(logical_or)	 scan(logical_or,...)
reshape(shape)	reshape(shape)	shape metadata + iterator view truncation	shape metadata update (lazy length/shape change)
cycle(shape)	cycle(shape)	thrustx::make_cycle_iterator	PermutationIterator(idx % N) (reshape when shrinking)
repeat(n) (scalar)	repeat(n)	thrust::make_constant_iterator	 constant(front(),n)
replicate(int n)	replicate(n)	thrustx::make_replicate_iterator	TransformIterator(idx//n) + PermutationIterator
replicate(mask)	replicate(mask)	materialized scatter + inclusive scan + permutation copy	 cupy.repeat (no scatter_if in cuda.compute)
cross(other)	cross(other)	 replicate + cycle + pairs	 replicate + gather(mod) + ZipIterator
outer(other, op)	outer(other, op)	thrustx::make_outer_iterator	Transform/Permutation/Zip iterator
pairs(other)	pairs(other)	zip + pair transform iterator	iterators.ZipIterator
enumerate()	enumerate()	 pairs(range(size))	ZipIterator(values, CountingIterator)
transpose()	transpose()	counting + transpose-index transform + permutation iterator	TransformIterator(idx) + PermutationIterator
rev()	rev()	thrust::make_reverse_iterator	iterators.ReverseIterator (with collect in some paths)
append(v)	append(v)	thrustx::make_append_iterator	index transform + PermutationIterator + TransformIterator
prepend(v)	prepend(v)	thrustx::make_prepend_iterator	index transform + PermutationIterator + TransformIterator